Multi-Parametric Approach to Investigate the Pre-Seismic Anomalies of M_w>6.0 using GNSS and Remote Sensing Satellites. A Case Study of Four Earthquakes from Mid Latitude Regions

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A thesis submitted to the
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I take full responsibility for the research work conducted during the MS Thesis titled “**Multi-Parametric Approach to Investigate the Pre-Seismic Anomalies of Mw>6.0 using GNSS and Remote Sensing Satellites. A Case Study of Four Earthquakes from Mid Latitude Regions”**. I solemnly declare that the research and development work presented in the MS Thesis is done solemnly by me with no significant help from any other person; however, small help wherever taken is duly acknowledged. I have also written the thesis by myself. Moreover, I have not presented this thesis previously to any other degree-awarding institution within Pakistan or abroad.

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Dr. Munawar Shah

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ABSTRACT

The recent advances in space-based ionospheric and atmospheric precursors of earthquakes aid in the development of a lithospheric atmospheric ionospheric coupling system. This study analyzes atmospheric and ionospheric anomalies associated with four different earthquakes of Mw>6.0 mid-latitude region (El Salvador, Afghanistan and Tajikistan). The atmospheric precursors are searched in humidity, Aerosol Optical Depth (AOD), and soil moisture from Giovanni (A Web interface to analyze NASA's gridded data). Furthermore, ionospheric anomalies in the form of total electron content were analyzed using GNSS-TEC. Pre atmospheric anomalies are observed before all three earthquakes during a 5- 50-day window beyond the upper bounds. Similarly, pre-ionospheric anomalies are also investigated before the impending earthquake. The temporal TEC observations retrieved from the International GNSS Service (IGS) stations of Managua and Guatemala cities, laying within the earthquake preparation area (EPA), showed a positive ionospheric response from 2 to 3 TECu from the corresponding upper bound of about 2-6 days before the EQ. The variations are found based on the confidence interval the of median and associated interquartile range for 10 days before and 5 days after the mainshock. The bounds are calculated from a continuous analysis of 10 days before and after the observed day. On the other hand, post atmospheric anomalies are only observed after Mw 7.0, the Afghanistan earthquake within 5 days. the overall analysis shows that pre atmospheric anomalies can be monitored more frequently than post atmospheric anomalies; however, more analyses need to clear the lithosphere atmosphere ionosphere coupling. There is a

lack of possible precursors of earthquakes but the need for more analysis can certainly validate the phenomenon.

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LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite Systems
PNT	Position, Navigation, and Time
USA	United States of America
LEO	Low Earth Orbit
MEO	Medium Earth Orbit
RNSS	Regional Navigation Satellite Systems
QZSS	Quasi-Zenith Satellite System
FOC	Full Operational Capability
FDMA	Frequency-division Multiple Access
CDMA	Code-division Multiple Access
EU	European Nation
GEO	Geosynchronous Earth Orbit
BDS	BeiDou Navigation Satellite System
IGSO	Inclined Geosynchronous Earth Orbit
IRNSS	Indian Regional Navigation Satellite System
SBAS	Space-Based Augmentation Systems
GNSS-R	Global Navigation Satellite System Reflectometry
M_w	Magnitude
UV	Ultraviolet
EMR	Electromagnetic Radiation
CME	Coronal Mass Ejection
UT	Universal Time
LAIC	Lithosphere-Atmosphere-Ionosphere Coupling
AOD	Aerosol Optical Depth
USGS	United States Geological Survey
DEM	Digital Elevation Model
NASA	National Aeronautics and Space Administration
OLR	Outgoing Longwave Radiation

1. INTRODUCTION

1.1. Global Navigation Satellite Systems

1.1.1. Introduction

Global Navigation Satellite Systems (GNSS) is a broad term used for satellite-based positioning that offers navigation solutions to users across the planet. We always require navigation to travel from one place to another and so forth. During the medieval era, navigation at oceans and seas was well developed. Likewise, troops on land require accurate mapping to cover huge areas. Similarly, precise timing and timekeeping have also remained a concern for centuries.

However, with the advancement of the latest technologies and recent development in the area of science, the issues regarding position, navigation, and time (PNT) are easy to interpret. Nowadays, problems like precise positioning, accurate timing, and navigation solutions can be provided by satellite technologies. Hence today, due to the development in satellite technology, we are observing multiple systems for satellite-based positioning, navigation, and timing that are orbiting around the Earth continuously. During the late 1950s, Russia launched the first satellite Sputnik into the outer space of our planet, but the first satellite that provides navigation solutions was the Transit satellite mission which was launched in October 1959 by the United States of America (USA) (Fig. 1.1). Since the inception and launching of the Transit mission, a new era of PNT has begun. The Transit mission operated from the Low Earth Orbit (LEO) and remained operational for 32 years while transmitting two different frequencies i.e., 149.99 MHz and 399.97 MHz [1]. Transit the mission was publicly available in 1967, providing a PNT solution to its users.

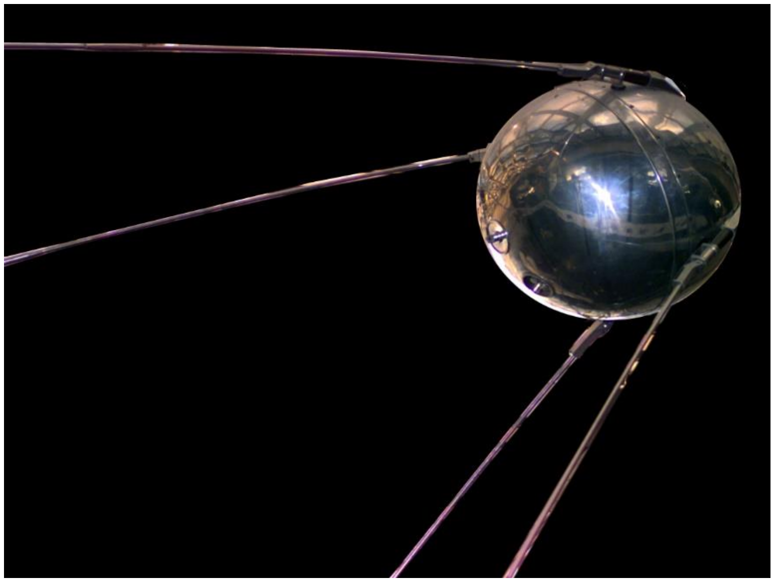

Fig. 1.1. Replica of Sputnik Satellite [Courtesy: NASA]

Later in 1974, Russia also had its first navigation-based satellite known as Cicada or Tsikada. Tsikada satellite mission also operates from LEO and transmits the same frequency as Transit had [1]. There are a few limitations with the Transit satellite mission i.e., only one satellite was visible at a given time due to this precise positioning jeopardized. So, the entire system is required to upgrade. Therefore, a new horizon of GNSS began in the early 1990s. Currently, four GNSS constellations (GPS, GLONASS, Galileo, and BeiDou) are in service. In the next section, we will discuss the GNSS constellation one by one.

1.2. Global Navigation Satellite Systems

As mentioned in the previous section, Cicada and Transit operated in LEO and only one satellite is available at a given time which made these satellite missions outdated in the late 20th century. Hence authorities identified those limitations and purpose of a completely new navigation system that will provide complete PNT solutions. As a result, in the early 90’s with the forthcoming Global Positioning System (GPS) and Global’naya Navigatsionnaya Sputnikova Sistema (GLONASS), Transit and Cicada satellite missions faded out as GPS and GLONASS provided more than 3 satellites at a given time. Unlike Transit and Tsikada, GPS and GLONASS operated in Medium Earth Orbit (MEO) and provide better accuracy as compared to their predecessors. The other great advantage of GNSS over their predecessors is that they can provide real-time to their users which were missing from earlier predecessors. As in the scenarios of Transit and Tsikada, the users of those satellite missions had to wait for the satellite to appear on the horizon to avail of services [1].

GNSS encompasses different segments like space Segment, ground Segment, and user Segment:

- **Space Segment:** Space segments consist of a constellation of satellites circling around Earth at an elevation of ~20000km from the surface of the Earth. Space segment also comprises inclination of orbit and keeping the satellite in its specific orbit.
- **Ground Segment:** The ground segment is also known as the control segment. It involves a master control station, locally monitoring stations and data uploading stations. The duty of this segment is to keep the strength of the satellite systems.
- **User Segment:** The user segment consists of receivers, antennas, different GNSS base appliances, and devices used to calculate the location, speed, and

time of the GNSS consumer (Fig. 1.2). This GNSS equipment varies from mobile phones to handheld GNSS base receivers used by users, to specialized receivers used in high-end mapping and surveying applications.

Currently, four GNSS constellations are providing high-level efficiency in position information to their users. GPS from the USA, GLONASS from Russia, Galileo from the European agency, and BeiDou (COMPASS) satellite navigation is from China. GNSS constellations transmit signals from space, which travel from the Earth's atmosphere and

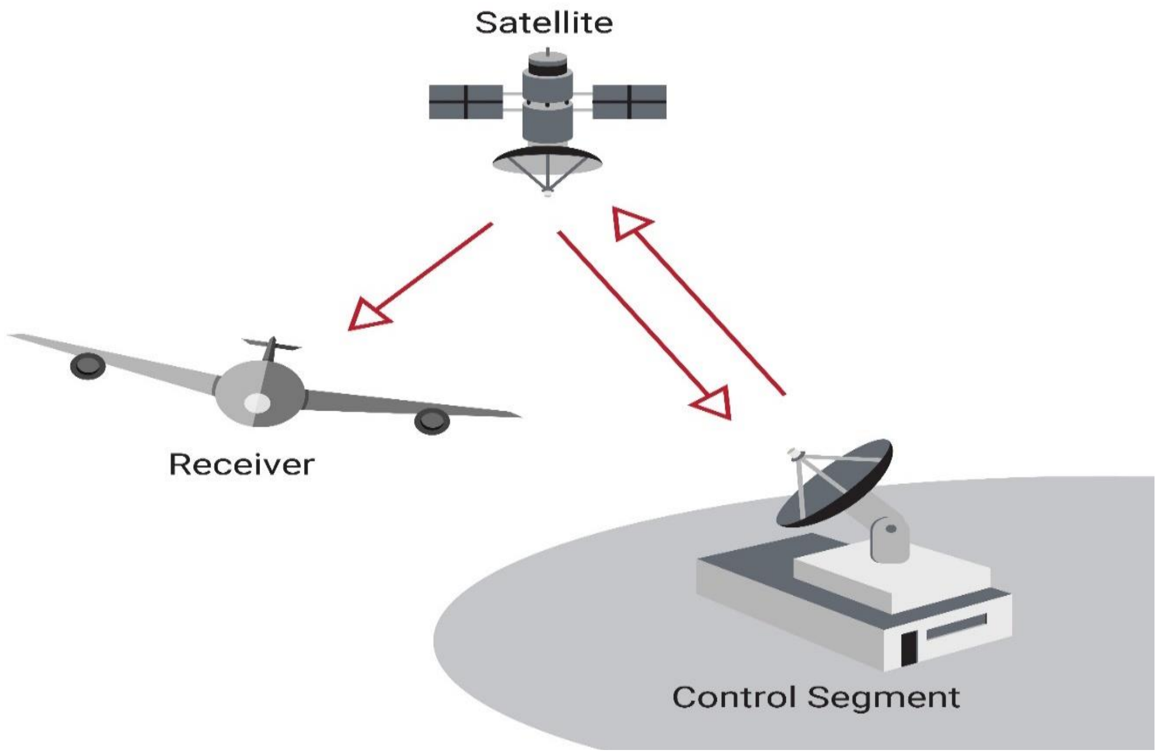

Fig. 1.2. Different GNSS Segments [Courtesy: VectorNav]

are received at their respective ground stations, where these signals further convert and provide information on position, velocity, and time.

All navigation satellite systems regardless of global or regional are working on the principle of trilateration. In the trilateration process, navigation satellite systems are used to determine the position of the user. In this process, the receiver position is determined by 3 different satellites that are available at a given time. Whereas, another satellite can be used for the accurate calculation of time. Fig. 1.3 depicts the illustration of the trilateration process.

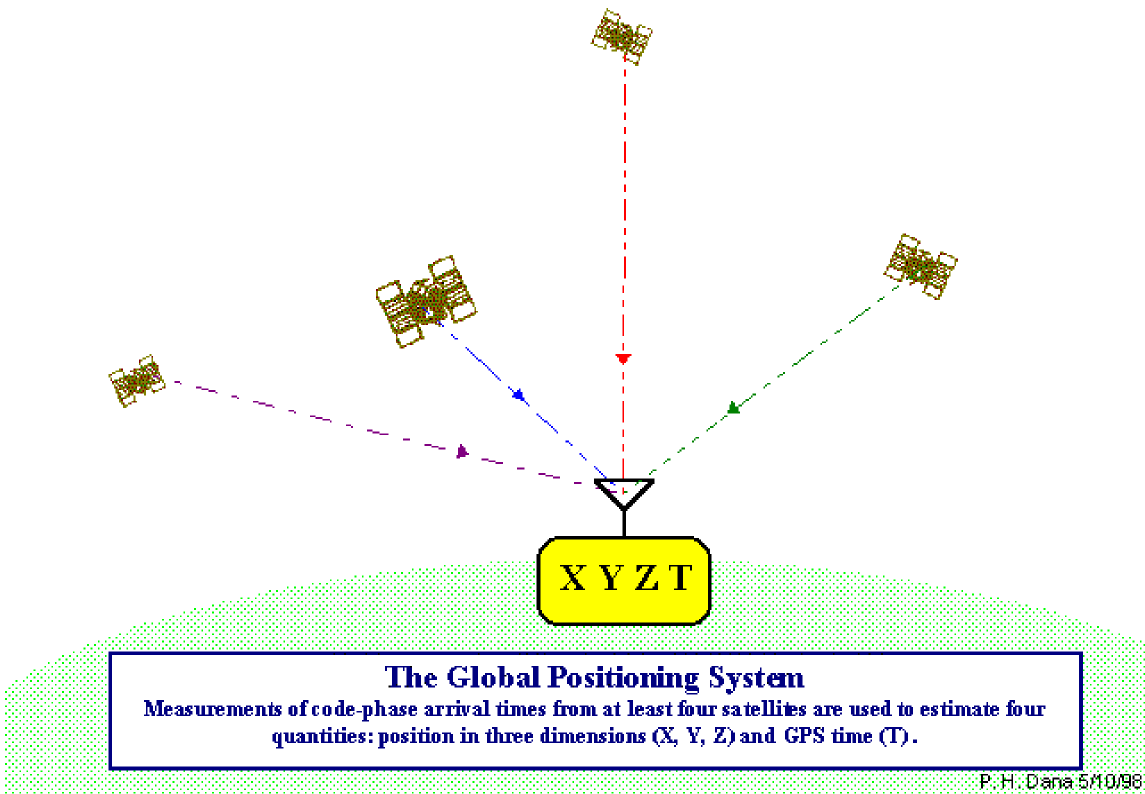

Fig. 1.3. Trilateration of GNSS process [Courtesy: University of Regina]

Apart from GNSS several Regional Based Navigation Systems (RNSS) missions are also operating on the principle of GNSS. These RNSS missions are Quasi-Zenith Satellite System (QZSS) from Japan and Indian Regional Satellite Navigation System (IRNSS) by India which is also known as Navigation for Indian Constellation (NavIC). Table 1.1 shows the information regarding GNSS and RNSS constellations.

1.2.1. Evolution of Global Navigation Satellite Systems

The inception and success of Transit opened new doors for scientists and engineers to work on space-based satellite navigation systems. Therefore, after the Transit satellite mission's success, USA-based navigation was developed. Initially, the project was under the name of NAVSTAR but after some time they opted for the name GPS for their upcoming satellite navigation venture. Under this project, they successfully launched the first satellite orbit around the Earth on February 22, 1978. Initially, the constellation was designed for MEO with 24 satellite vehicles orbiting around the Earth at an altitude of above ~20,000km from the surface of the Earth. The project was completed in December 1993 with full operational capability (FOC) to be declared in 1995. GPS was initially purposed to operate under two frequencies (i.e., L1 and L2) [2]. L1 frequency of GPS is designated on the civilian to have free access along with coarse/acquisition (C/A) code and a navigation data message. Other than that, another signal is modulated on both L1 and L2 frequencies known as the P(Y) code, which is only accessible to authorized personnel. Later with the advancement of satellite technology and the modernization of GPS block-II to GPS-III block (which began in 2005); L5 and L2C signals were also made available for civilian use. Subsequently, military M-Code was also incorporated to improve jamming and spoofing attacks on L1 and L2 signals. As GPS signals travel from space to the upper and then in the lower atmosphere, these signals may incorporate many errors from the atmosphere medium. One such error was selective availability which can be incorporated into GPS signals for security reasons. However, in May 2000, the selective availability service withdrew by the US government to improve the reception of GPS signals and availability to their users.

Like GPS, the Soviet Union decided to launch their navigation satellite constellation therefore, on October 12, 1982, the Soviet Union launched their GNSS named GLONASS. GLONASS was initially launched in MEO having a constellation of 24 satellites. However, after the fall of the Soviet Union, Russia's economy suffers, causing great damage to its GLONASS project. Yet, after Russia's government support, the FOC was retained in late 2011 and maintained subsequently [3]. Later, at the beginning of the 21st century, Russia decided to modernize the GLONASS system, so they revolutionized the constellation by launching modernized satellites and configurations, aiming to provide better navigation solutions to its users. Initially, GLONASS uses Frequency-division Multiple Access (FDMA) for the transmission and receiving of the signals. But in the modernization program of the GLONASS, they FDMA to Code-division Multiple Access (CDMA). And on February 26, 2011, Russia launched its first modernized satellite GLONASS-K (Third Generation) which is using CDMA instead of FDMA.

The success of GPS and GLONASS intrigues many nations to work on their separate navigation systems. As a result, European Nations (EU) decided to launch their GNSS under the name Galileo. At first, two MEO satellites were launched on October 21, 2011, then on October 12, 2012, a set of four Galileo launched in MEO, and by the end of 2015, six additional Galileo satellites were to complete constellations. Furthermore, European Space Agency (ESA) planning to have 30 satellites in its constellation with 24 primary satellites and 6 satellites to remain reserve satellites in the constellation. The signal structure of Galileo is the same as that of GPS. The E1 of Galileo uses the same frequency bandwidth as that of is L1 of GPS [4]. A detailed comparison of navigation satellite systems is provided in Table 1.1.

The fourth GNSS constellation is the Chinese BeiDou Navigation Satellite System (BDS), which initially was designed to provide services for regional-based navigation. BeiDou project was started in 2003 with the launching of three satellites in the Geosynchronous Earth Orbit (GEO), then in 2007, after the launching of the fourth satellite in GEO, they had a constellation of 4 GEO satellites able to deliver the solution for the region based navigation. Later, by advancing their constellation, they managed to achieve BDS as GNSS status by having 7 GEO, 25 MEO, and 5 Inclined-Geosynchronous Earth Orbit (IGSO) satellites. The main objective of BeiDou is to provide designated services for authorized users to access highly precise accuracy from BeiDou and to provide free services for civilians with position, velocity, and timing accuracy of 10m/s, 0.2 m/s, and up to 50 nanoseconds respectively [5]. Currently, BDS has B1, B2, and B3 frequencies operational for its constellation.

1.2.2. Evolution for Regional Navigation Satellite Systems

Subsequently, after the success of GNSS, countries began to develop different regional navigation satellite systems to provide better services within their region. Indian Regional Navigation Satellite System or NavIC (IRNSS) from India and Quasi-Zenith Satellite System or Michibiki (QZSS) by Japan are the two RNSS that are providing regional coverage to their users. Japan comprises an uneven mountainous terrain-like region that requires higher accuracy from navigational satellites in their region to provide better services for their users. Hence there is a demand for their regional-based navigation, which can provide them with a better navigational solution. Therefore, on September 11, 2010, Japan launched its first satellite under the QZSS project in a highly elliptical orbit with a 70^{o} orbit inclination. So far, 4 satellites have been launched under the QZSS mission that provides better accuracy and navigation solutions to its users. QZSS uses L1, L2, L5, and E6 bandwidth signal frequency for the transmission and receiving of signals.

Table 1.1. Information regarding Navigational Satellite Systems

System	No. of Satellites	Orbits	Frequency [MHz]	Initial Service	Country	Coverage
GPS	24	MEO	L1, 1575.42 L2, 1227.60 L5, 1176.45	Dec 1993	US	Global
GLONASS	24	MEO	L1, 1602.00 L2, 1246.00 L3, 1202.02	Sep 1993	Russia	Global
Galileo	30	MEO	E1, 1575.42 E5a, 1176.45 E5b, 1207.14 E6, 1278.75	2016-17	Europe	Global
BeiDou	35	MEO, GEO, IGSO	B1, 1561.09 B2, 1207.14 B3, 1268.52	Dec 2012	China	Global
QZSS	4	GEO, IGSO	L1, 1575.42 L2, 1227.60 L5, 1176.45 E6, 1278.75	2018	Japan	East Asia Oceania region
NavIC	7	GEO, IGSO	L5, 1176.45 S, 2492.028	2016	India	Regional

Other than QZSS regional navigation system, IRNSS is also operational to provide services for its users. IRNSS operates with two frequency bands i.e., S-band and L5 band [6]. IRNSS launched 7 seven satellites in geostationary orbit which aims to provide its services within the range of 1500km. The expected accuracy for IRNSS is 20m in the Indian Ocean and 10m accuracy on land [7]. Apart from these two RNSS, other countries are also planning to launch their RNSS.

1.3. GNSS Augmentation Systems

Apart from RNSS, augmentations systems or space-based augmentation systems (SBAS) are also launched to assist GNSS in providing better accuracy and navigation solutions for GNSS users. Countries use GNSS and integrate different SBAS with them to achieve better positioning and navigation services.

SBAS satellites are generally launched in GEO to provide better navigation, position accuracy and reliability, and better correction to their users. Currently, four SBAS missions are operating and are orbiting around Earth to provide their services to its users. So far all augmentation systems are operating on the GPS frequency band [8]. Table 1.2. illustrates the information regarding all launched augmentation systems. Other than four launched missions, other countries are planning to launch their SBAS in the future. Henceforth, Chinese Beidou SBAS (BDSBAS), Korea Augmentation Satellite System (KASS), SBAS for African and Indian Ocean (A-SBAS), Australia and New Zealand's; Southern Positioning Augmentation Network (SPAN) is currently in the development phase [9]. There are Continuously Operating Reference Stations (CORS) are also installed by many countries that help them to achieve better GNSS accuracy and precision for their users. CORS network uses post-processed data and applies corrections to that data. The application of CORS ranges from spatial mapping to surveying and mapping [10].

Table 1.2. Information regarding Augmentation Systems

S. No.	Name of Augmentation System	Origin of Country	Coverage Area	Signal in Space Range Error	Ionospheric Correction Accuracy
1	Wide Area Augmentation System (WAAS)	United States of America	North America	0.954m	0.505m
2	European Geostationary Navigation Overlay Services (EGNOS)	European Union	Europe, Africa, and Partially Canada	0.645m	0.491m
3	GPS Aided GEO Augmentation Navigation (GAGAN)	India	Indian region, Africa and Australia	0.795m	0.858m
4	Multi-functional Satellite Augmentation System (MSAS)	Japan	Japan region	1.931m	1.325m
5	System for Differential Corrections and Monitoring (SDCM)	Russia	Russian region and partial Europe	0.650m	0.523m

1.4. Interferences in GNSS

GNSS constellations have an altitude of ~20,000km and above the center of the Earth, therefore, when the signal transmits from the satellite it has to deal with multiple inferences before reaching the GNSS receivers. The interferences cause errors in the signals which affect the strength of the signals and hence vulnerability can be observed in those signals. Generally, interferences in GNSS can be categorized as follows:

i) Deliberate Interferences

ii) Undeliberate Interferences

1.4.1. Deliberate Interferences

Deliberate interferences in GNSS are also known as intentional interferences. In these sorts of interferences, an error intentionally implants in GNSS signals causing the signal to disrupt or delay services. Generally, selective availability, signal jamming and spoofing are three types of such inferences that are considered as deliberate interferences.

In selective availability, due to security concerns, the USA government deliberately reduced the accuracy of satellite clock corrections. Out of four GNSS constellations, only GPS had the feature of selective availability therefore, to make GPS more efficient and accessible to users GPS has to withdraw this feature from its service. Eventually, on May 2, 2000, selective availability is permanently withdrawn by the USA government to make GPS more efficient for the consumer. After the launching of block III GPS satellites, GPS does not support this feature anymore.

Table 1.3. Error sources along with their ranges

Error Source	Error Range
Satellite Clock Error	±2 m
Satellite Orbit Error	±2.5 m
Ionospheric Error	±5 m
Tropospheric Error	±0.5 m
Receiver Noise	±0.3 m
Multipath	±1 m

Signal jamming is one of the most common causes of intentional error. In signal jamming, a fake signal is deliberately implanted in the original GNSS signal causing the receiver unable to track the signal. Therefore, by the use of multi-constellation receivers, these sorts of errors can be avoided.

In GNSS signal spoofing, a replica or a mock GNSS signal can be used to make a receiver believe it is at a false location. GNSS signal spoofing is considered one of the most dangerous sorts of interference than any other interference as it is hard to detect. To overcome GNSS signal spoofing, the detection of arrival technique is used. In this technique, a GNSS receiver predicts the arrival time of the signal hence it can use this information to overcome the GNSS signal spoofing error.

1.4.2. Undeliberate Interferences

Undeliberate interferences are also known as unintentional interferences or random interferences. Many phenomena or factors can cause these sorts of errors in GNSS signals. Table 1.3 categorizes error sources along with their impact on GNSS signal accuracy.

The primary objective of every GNSS constellation is accurate PNT. Therefore, all the GNSS satellites of equipped with a very precise atomic clock. Still, the clock on the user end drifts from the GNSS time, which can cause disturbance as a result accuracy of the GNSS consumer can be compromised and an error of ±2 m can be observed. Another type of disturbance that can cause an error of up to 2.5 m is observed and this type of error is known as satellite orbit error. The generation of this type of error occurs when a satellite changes its orbit. Even though the ground station updated the ephemeris via correction there remains a slight error in the orbit of a satellite. Therefore, this sort of error is known as satellite orbit error.

As GNSS constellations are orbiting around Earth from above ~20,000km from the center of the Earth, therefore, these GNSS signals have to propagate from the outer space into the atmosphere of the Earth, while traveling into the Earth’s atmosphere these signals have to deal with many interferences that can cause major delays in GNSS signals. As Earth has its atmosphere, hence Earth’s atmosphere is divided into several layers depending on the specifications and altitude of each layer. As GNSS signals entered Earth’s atmosphere it has to propagate through the ionospheric layer of the atmosphere. The ionospheric layer of the Earth’s atmosphere is a highly charged layer due to the presence of free electrons, therefore, these free cause the GNSS signals to deflect and diffract from their original path, and hence as a result of this interference, an error of up 5m can be observed. Likewise, while entering the tropospheric layer of the atmosphere (also known as the lower atmospheric layer) GNSS signals interact with water molecules that can cause refraction in GNSS signals [11]. This reflection of GNSS signals can also be one of the reasons to cause delays in GNSS signals.

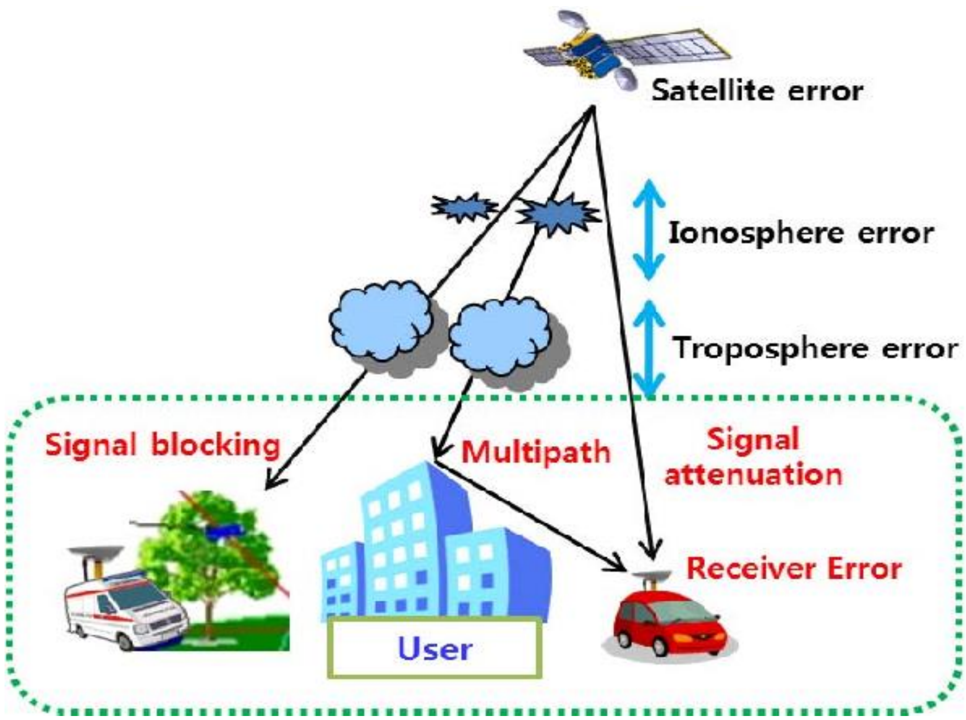

Fig. 1.4. Illustration of GNSS errors [Courtesy: D. Won et al., 2015]

After passing the lower atmosphere GNSS signals enters the surface level of the Earth. Before reaching the receivers, they have to deal with several parameters that can affect the quality of the signal. This sort of error in the GNSS signal is known as multipath error or receiver noise error. Cheap GNSS receiver hardware or software is responsible for the receiver noise error, while multipath interferences are generated due to the reflection or refraction of GNSS signals with any object before reaching the receiver.

1.5. Applications of GNSS

The general purpose of the GNSS is to provide PNT to its users, but it can also provide a wide range of applications from defense to farming, timing to aviation, and from metrology to marine, etc. Fig 1.5 illustrates different applications of GNSS. GNSS reflectometry (GNSS-R) can provide applications like precision in altimeter, wave height, and wind speed in the oceanography, ice monitoring, deformation of land, and soil moistening system. GNSS-R can benefit the agriculture sector by precisely identifying and monitoring the soil moisture content which can be benefitted farmers to work more efficiently [12]. Another example of the application of GNSS is surveying and mapping. As GNSS can provide the exact latitude and longitude of the respective area that can help surveyors to work more precisely and efficiently as compared to conventional mapping techniques [13].

Fig. 1.5. Illustration of GNSS Applications

GNSS can also provide timing applications to its users as GNSS constellations are equipped with highly accurate atomic clocks that can be advantageous for achieving better timing accuracy. Modern-day mobile phones are also assisted by GNSS as they are using GNSS timing navigation accuracy. All sorts of modern way communications require GNSS constellations assistance to provide better accuracy and navigation solutions. Apart from these, GNSS also assist drones and autonomous vehicles as all those systems require geo-referenced information and terrain information to maneuver themselves, therefore, these systems may use GNSS constellations for better accuracy and navigation [14], [15] and [16].

GNSS signals can travel a long distance from the outer to the Earth's atmosphere and the ground level, hence they can provide great insight into the atmosphere and can predict the atmosphere. Natural disasters like seismic effects, cyclones, typhoons, volcanic eruptions, and land deformation can have serious effects on Earth's atmosphere. With help of GNSS signals, researchers can predict the forthcoming of natural hazards and may be able to generate the early alarm system to help the authorities and minimize the damage.

1.6. Motivation and Objective

This study is regarding the analysis of seismic-induced atmospheric precursors that cause damage in the region of Afghanistan and Tajikistan region. The key objective of this study is to inspect and analyze the pre and post-seismic atmospheric anomalies and their relationship with the help of GNSS-R and other remote sensing satellites. The study focuses on the Earth's atmosphere and its behavior during the seismic activity associated with the three EQs in the Afghanistan and Tajikistan region from 2005-2011 having magnitudes (M_w) of greater than 6. In this study, we are using GNSS-R to analyze the soil moisture content during the seismogenic period for the EQs having M_w greater than 6.0.

In the next chapter, the author gives insight regarding the Earth's atmosphere and the behavior of layers of the Earth's atmosphere. Moreover, atmospheric behavior during seismic-related activities is also examined and analyzed the relationship between the atmosphere and seismic-induced activity.

So far, a brief history of satellite technology and GNSS along with their errors and application are discussed in this chapter, while in chapter 2, we will further discuss the Earth's atmosphere and its behavior during the seismic related activity and how satellite technology can help predict and analyze the seismic related anomalies. Data and methodology will be discussed in chapter 3, whereas, results and discussion on results will be discussed in chapter 4. In the end, future work and recommendations will be discussed in chapter 5.

2. Earth Atmosphere and Seismic Waves

2.1. Earth Atmosphere

Earth has an atmosphere that comprises different gases ions and molecules that are surrounded within a planet due to the gravity of Earth. As mentioned, the atmosphere of the Earth comprises gases particles, and molecules and has a series of layers covering the Earth. These layers are also known as atmospheric layers. Fig. 2.1., shows different layers of the Earth's atmosphere. Atmospheric layers of the Earth are categorized as follows:

i) Troposphere ii) Stratosphere iii) Mesosphere iv) Thermosphere

These Earth's atmospheric layers will be discussed in the next section:

2.1.1. Troposphere

The troposphere is the lowest part of the Earth's atmospheric layer. This layer is responsible for all the weather activities like rain, thunderlight, snow, clouds, etc. This layer ranges from the surface of the Earth and extends up to 10km in altitude. In the tropospheric layer, temperature decreases with the increase in altitude. As gases are the primary constituent of the Earth's atmosphere, therefore, the most abundant gas in the troposphere is Nitrogen, which makes the 78% of the atmosphere. On the contrary, Argon makes up to 1% of the Earth's atmosphere while Oxygen makes up to 21% of the Earth's atmosphere. At the end of the troposphere or the upper layer of the troposphere, tropopause exists. It is considered as the boundary layer for the troposphere, as after the troposphere, the stratosphere starts. The tropopause layers range up to 18km in the equatorial region, whereas, in the polar region this layer appears to fall up to 8km in altitude [16].

Temperature decreases concerning altitude in the troposphere whereas, in the tropopause, temperature rather remains constant or has a low affect on this layer.

2.1.2. Stratosphere

The stratosphere is the second layer of the Earth's atmosphere after the troposphere. This layer makes up to 50km in altitude of the Earth's atmosphere and has a concentration of ozone particles. Therefore, the stratospheric layer is also known as the ozone layer of the atmosphere. The stratosphere is also responsible to prevent the ultraviolet (UV) and x-rays that are coming from the Sun, as this layer absorbs these radiations due to the presence of the ozone particles. In the stratospheric layer, the temperature rises with the increase in altitude. At the top of the stratosphere, stratopause exists which acts as a boundary between stratosphere and mesosphere.

2.1.3. Mesosphere

The mesosphere is the third layer of the atmospheric layer of the Earth after the end of the stratosphere and stratopause. In the mesosphere, as with the scenario of the troposphere, temperature decreases with the increase in altitude. The mesospheric layer of the atmosphere starts from ~50km above the surface of the Earth till 80km in altitude. The mesosphere has abundant concentrations of ozone gas particles. The mesosphere is responsible for the happening of the southern and northern lighting also known as auroras. At the top of an altitude, mesopause exists which is also the coolest region in the atmosphere of the Earth.

After the mesosphere and mesopause, the thermosphere layer of the atmosphere exists which is considered the fourth layer of the atmosphere.

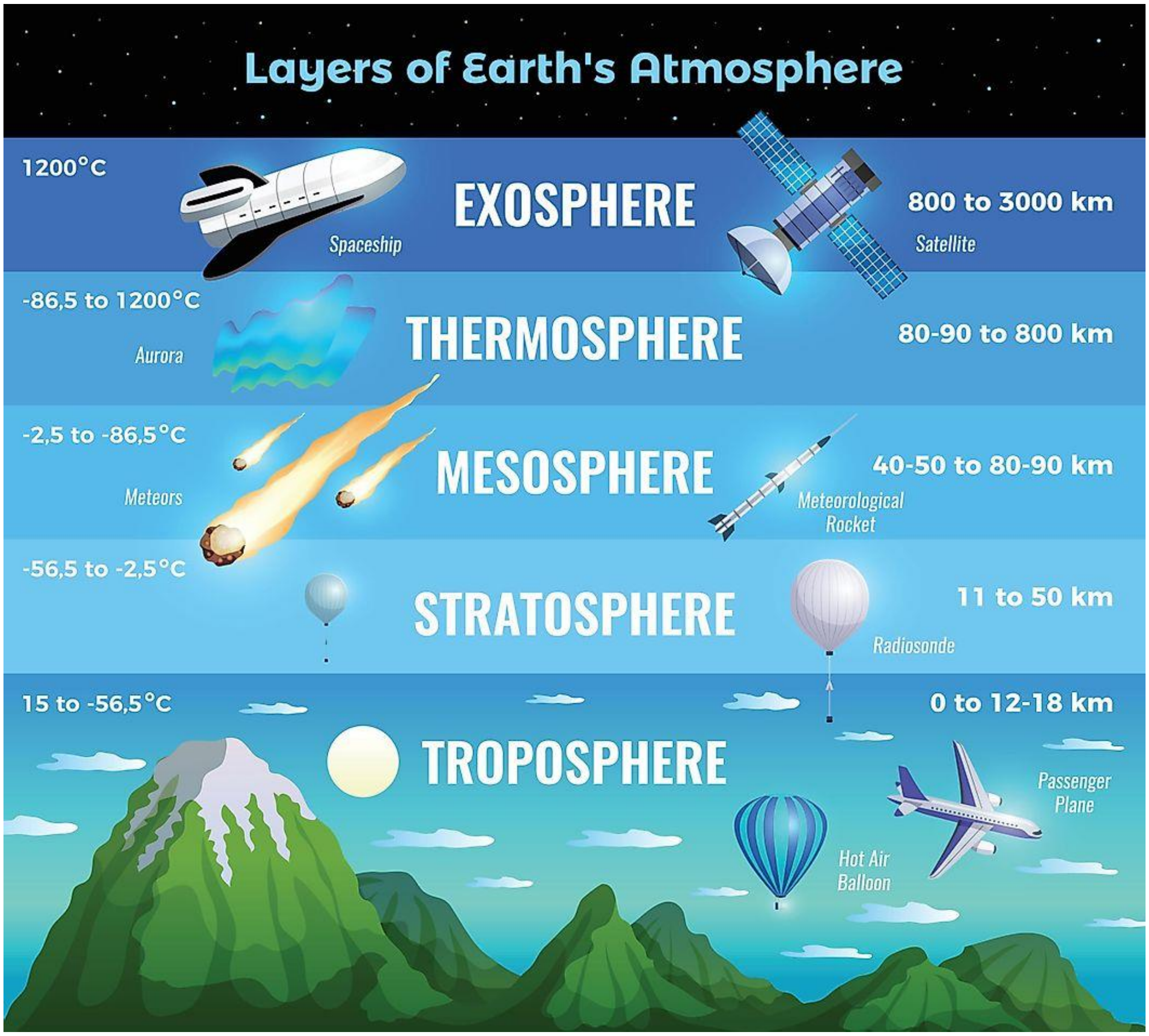

Fig. 2.1. Illustrates different layers of the Earth's atmosphere [Courtesy: World Atlas]

2.1.4. Thermosphere

The thermosphere is the fourth layer of the Earth's atmosphere which starts after the end of the mesosphere and before the start of the exosphere. In the thermosphere, the presence of free electrons and molecules in this layer contributes to the maximum portion in the ionosphere layer. As the presence of charged particles and molecules, the UV from the Sun causes photoionization of the molecules in this layer. The thermosphere layer starts from 80km in altitude from the surface of the Earth. This layer has contact with the Solar energetic particles from the Sun that causes the rise of temperature in this layer. The temperature in this layer can rise to 2000^{o}C making this layer the most heated layer of the Earth's atmosphere.

The region of the atmospheric layer between the stratosphere and exosphere is also known as the ionosphere layer as in this region free electrons and particles are abundant.

2.2. Ionosphere

The region between the stratosphere and exosphere presents free electrons; therefore, this region is generally known as the ionosphere or ionospheric region of the atmosphere. In this region, the Sun's electromagnetic radiation (EMR) causes the photoionization process in the ionosphere which results in the abundance of the free electrons in this layer. In the ionosphere region, due to the interaction of EMR, this region divides into several sub-layers, and each sub-layer has its significance. Fig 2.2 illustrates different sub-layers of the ionosphere. The ionospheric layer has a unique implication in the propagation of radio signals. Due to the presence of free electrons in the ionosphere, signals transmitted from space to the Earth or from the Earth to space observe various perturbations while passing from the ionosphere. In the alone GNSS, the ionosphere can cause significant disturbances in the accuracy of the GNSS signals. Several models are introduced to overcome the issues regarding the perturbations in the ionospheric layer of the Earth and for the smooth processing of GNSS signals [17], [18], and [19]. Solar activity is directly proportional to the behavior of the ionosphere as the intensity of the solar activity will have a direct impact on the photoionization process in the ionosphere [20]. The range of the ionosphere in the

Earth's atmosphere starts from ~60km to ~1000km in altitude. The sub-layers of the Earth ionosphere are named D, E, F1, and F2, which lie in the different regions in the ionosphere.

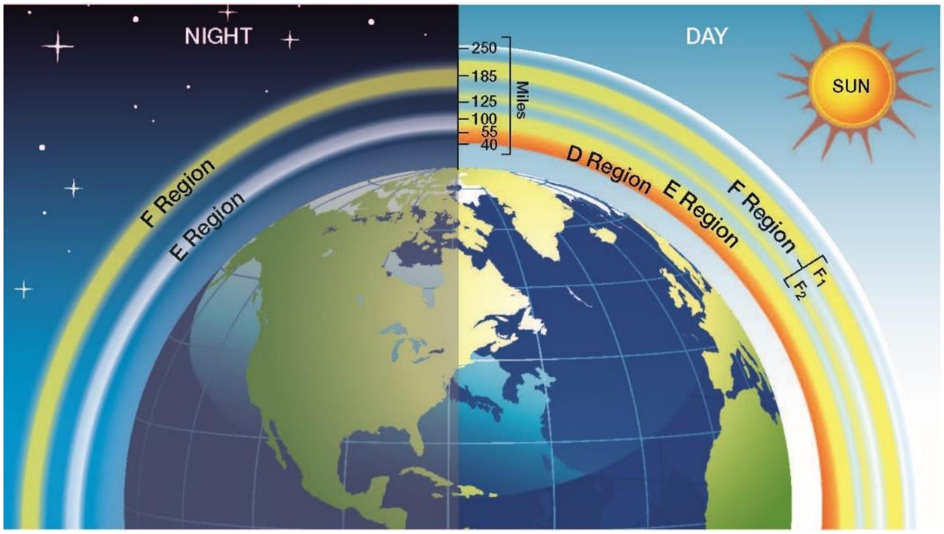

Fig. 2.2. Different Regions of Ionosphere [Courtesy: NASA]

In the D-region of the ionosphere, the EMR from the Sun causes ionization in molecular nitrogen, nitric oxide, and oxygen [21]. D-region of the ionosphere starts from ~60km in altitude and extends up to ~85km in height. D-region is considered as the lowest region of the ionosphere, and during the daytime, due to the concentration of free electrons, this layer merged with the E-layer of the ionosphere. Above D-region, the E-region of the ionospheric layer of the Earth exists that ranges from ~85km to ~150km in altitude. E-region is relatively stable as compared with D-region, as E-region covers the particular region in the ionosphere during the day and nighttime. F-region of the ionosphere comes after the E-region starting from ~150 km in altitude and extends up to ~1000 km during the daytime. Hence this region is considered the uppermost region of the ionosphere. Due to the maximum photoionization process during the daytime, F-region splits into two separate layers (i.e., F1 and F2) [22]. On the contrary, during the nighttime, this region merges and forms into a single layer which is known as the F-region of the ionosphere.

Perturbations in the ionosphere can cause severe problems for the signals that are passing from the ionosphere. Due to the presence of free electrons in this layer, the Sun can cause significant variations in this region. Therefore, it is vital to discuss how Sun can cause disturbances in the ionosphere.

2.3. Solar Disturbances in the Ionosphere

In section 2.2., various regions of the ionosphere are discussed. In this section, the Sun effect of the ionosphere will be discussed. As the Sun is continuously transferring its energy in outer space, therefore, this energy also reaches the Earth and causes disturbances in the atmosphere. EMR from the Sun can reach into the Earth’s atmosphere with the speed of light and cause disturbances in the atmosphere and even cause disturbances into power lines or power grids.

In this section, different types of interferences and impacts from the Sun in the ionosphere will be discussed. Table 2.1 shows different solar and geomagnetic indices along with their intensity.

2.3.1. Solar Wind

The solar wind is the release of the charged particles from the Sun’s atmosphere into outer space (fig. 2.3). The charged particles that burst from the Sun’s atmosphere are composed of plasma particles or electrons and protons that tend to enter the atmosphere and cause disturbances. Coronal Mass Ejection (CME) and solar flares are responsible for the generation of solar wind and these generated solar winds tend to carry solar flares and CMEs to outer space. Strong solar winds might tend to change the magnetic field of the planet. In the scenario of the Earth, Earth has its magnetic field also known as "Van Allen Belts" that can trap the solar winds the divert them into outer space [23].

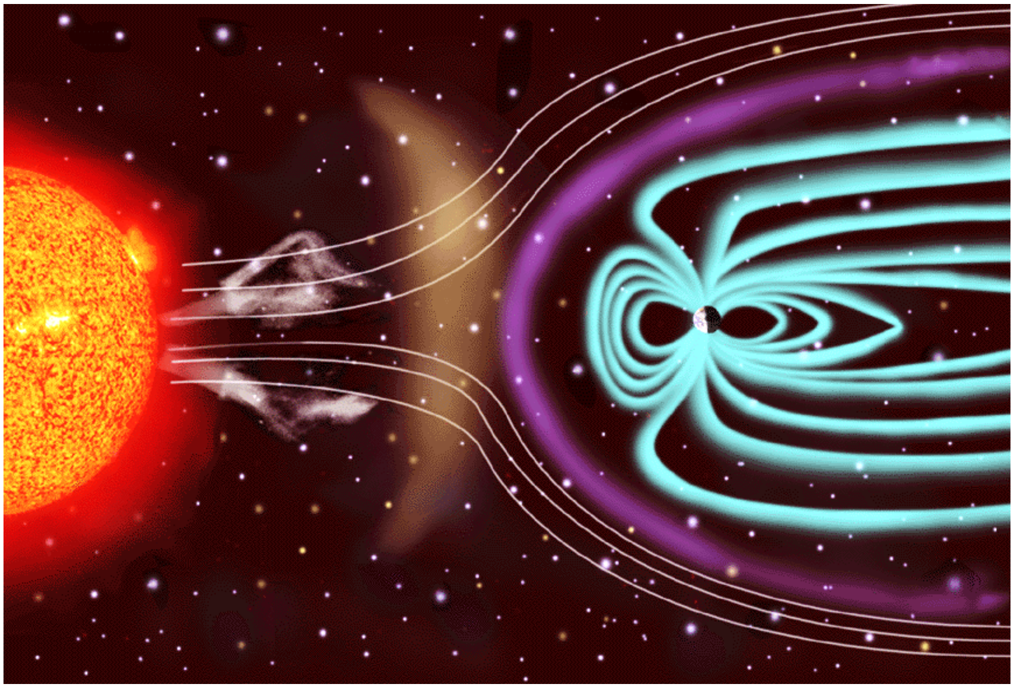

Fig. 2.3. Depiction of Solar Wind into the outer space [Courtesy: ESA]

Auroras are the common example of the interaction of solar wind with the Earth's atmosphere, as Earth's magnetic fields are relatively weaker from the poles therefore, the solar wind can penetrate the upper atmosphere and cause northern and southern lighting also known as Auroras in the polar region of the Earth.

2.3.2. Sunspot

Sunspots are formed due to the emission of the magma that erupts from the inner surface of the sub-atmosphere of the Sun are relatively cooler areas on the surface of the Sun. The temperature in sunspots areas dropped to ~4000k. Fig. 2.5. illustrates the sunspot on the surface of the Sun. CMEs or intense solar flares can cause the formation of the sunspot on the surface of the Sun.

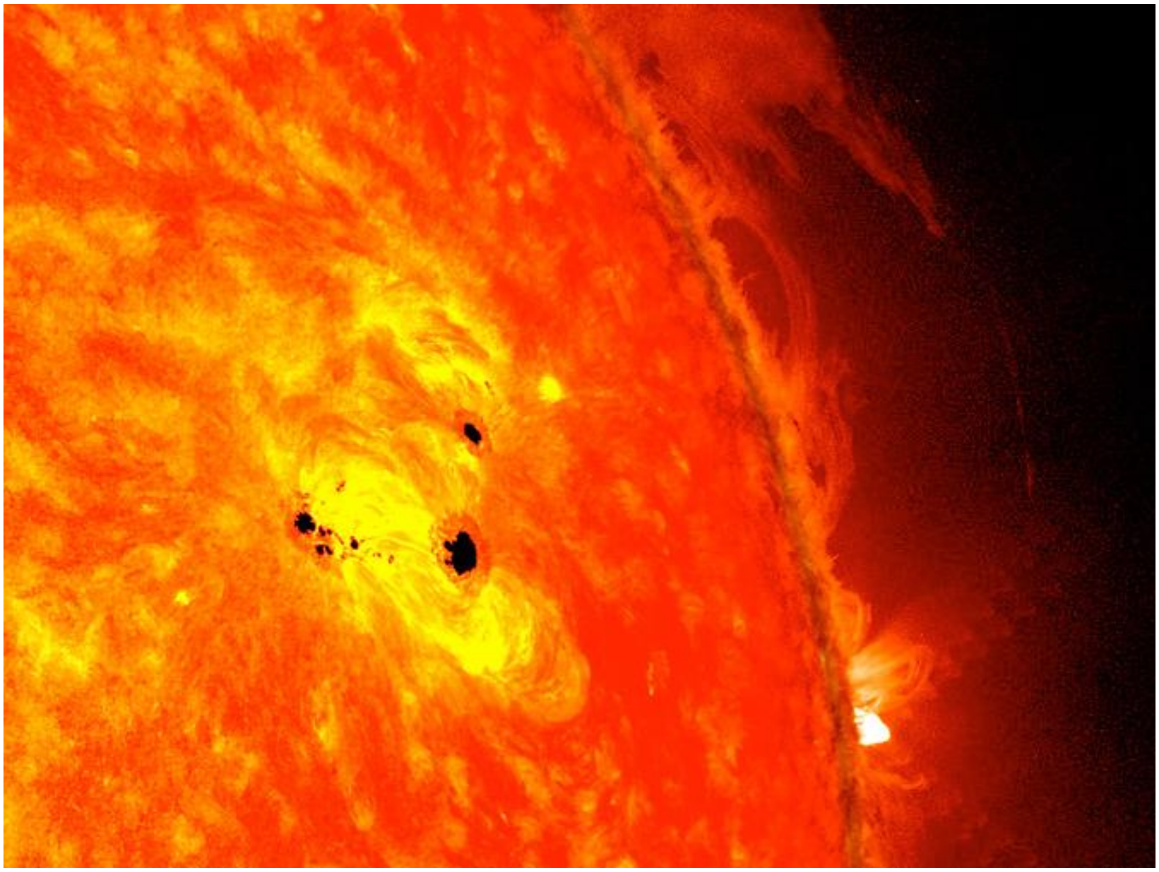

Fig. 2.4. Illustrates the sunspot on the surface of the Sun [Courtesy: NASA]

Like the Earth, Sun has its magnetic activity known as the solar cycle or solar magnetic activity. The solar magnetic activity is almost an 11-year solar cycle formed by the Sun. During this cycle, when the number of sunspots increases that phase is considered as solar maxima. After solar maxima, Sun's magnetic field reverses that reversal of magnetic field settles the Sun back to minimum storm formation. This phase of the minimum storm formation on the Sun is considered as solar minima. The relative sunspot number index is a guide to analyze or monitor the entire visible disk or surface of the Sun. The eruption of the sunspots may last for weeks or remains visible on the surface of the Sun for months [24].

2.3.3. CME and Solar Flares

Solar flares are the sudden and intense bursts of plasma radiation that releases from the magnetic energy associated with the sunspots. While CMEs are the massive bubble or cloud-like EMR that ejects from the Sun's surface into outer space. Fig. 2.5 shows the eruption of CMEs or solar flares along with the formation of auroras in the Earth's atmosphere. These solar flares and CMEs are responsible for the different space weather phenomena and their association with the Earth. Both the solar flares and CMEs tend to energize the solar particles and EMR that can cause disturbances in space weather and also in the ionospheric region of the Earth [25]. These interactions of solar particles and EMR with the Earth's atmosphere are also responsible for southern and northern lighting (also known as auroras). The eruption of the solar flares and CMEs is the reason for the appearance of the visible dark stains or spots observed on the photosphere (Visible surface of the Sun) Sun. As discussed in section 2.3.2, a weaker magnetic field on the surface of the Sun is responsible for the formation of the sunspots.

Fig. 2.5. The left panel illustrates the formation of Auroras and the right panel depicts the eruption of the Solar Flares and CME [Courtesy: NASA]

2.3.4. Geomagnetic Disturbances on the Earth

As discussed in the above section, Solar disturbances can cause severe disturbances in the atmosphere of the Earth. As the ionospheric region is the outer atmospheric region of the Earth's atmosphere, therefore this layer has a direct interaction with solar EMR. With the interaction of EMR from the Sun, the magnetic storm is formed in the ionosphere. These disturbances or magnetic storm in the ionosphere layer is known as geomagnetic storm or ionospheric storms. Table 2.1. shows geomagnetic storm indices (Ap, Kp, Dst, and AE) along with their severity on the ionosphere.

The intensity of geomagnetic storm disturbances is analyzed by the Ap index. If the value of the Ap index is exceeded by 50 then it is considered to be a strong planetary index (Ap index) [26]. Likewise, the Geomagnetic activity or impact of the solar storm on the magnetic field of the Earth is monitored by using the Kp index. Like Ap, Kp is also considered as a planetary index range on the scale from 0-9 and is observed in a given 3-hour of the universal time (UT) [27], [28].

Table 2.1. Classification of Geomagnetic Indices along with their impact

Geomagnetic Index	Phases				
	Quiet	Low	Moderate	Strong	Intense
Ap	≤ 20 nT	≥ 30 nT	≥ 50 nT	≥ 80 nT	≥ 100 nT
Kp	0	≤ 3	4	5-6	7-9
Dst	≥ -20 nT	≤ -30 nT	≤ -50 nT	≤ -100 nT	≤ -200 nT
AE	≤ 200 nT	≤ 400 nT	≤ 800 nT	≥ 1000 nT	≥ 2000 nT

Auroral Electrojets (AE) is also one of the phenomena occurring in the upper atmosphere of the Earth that are associated with the currents in the magnetosphere. AE is formed in the polar regions of the Earth's ionosphere due to weak magnetic fields in that region and severe weather conditions in the outer space of the Earth due to storm activities these AE's tends to shift towards the equator [29].

Another disturbance in the upper atmosphere of the Earth is known as the "Disturbance Strom Index" or Dst index. Through the Dst index, researchers measure the intensity of the ring current around the Earth's atmosphere. This ring current helps Earth from unwanted solar EMR by deflecting or trapping the solar EMR from entering into the Earth's lower atmosphere. The magnetic disturbance is associated with storm-related activity if the Dst index lowers from -50nT. This is also referred to as a weak Earth magnetic field as in this phase more solar particles penetrate Earth's atmosphere and cause disturbances in the atmosphere [30]. The classifications of different geomagnetic indices along with their intensity levels had explained by Gonzalez et al., [31] and illustrates in Table 2.1.

All these phenomena cause disturbances in the upper atmosphere of the Earth that can affect satellite communications and even power grids on the Earth. GNSS can also be affected by the disturbances in the atmosphere of the Earth. The propagation of the GNSS signals from the space to ground-based receivers will be discussed in the next section.

2.4. Impact of Ionospheric Disturbances on GNSS

GNSS constellations are continuously orbiting around Earth from ~20,000km above the surface level of the Earth. Hence the GNSS signals travel from space towards the ground-based receivers, therefore, GNSS signals have to deal with several disturbances that may cause efficiency delays in the propagation of the signals. The Ionospheric region contains free electrons which can refract and deviate GNSS signals from their original path resulting in the accuracy of errors in GNSS signals. Fig. 2.6., depicts the traversing of the GNSS signals from the ionosphere region. As mentioned earlier, the Sun can cause the photoionization process in the ionosphere causing the ionization process in this layer which results in causing the refraction of the GNSS signals and disturbing the TEC quantity in the ionosphere. The refraction in the ionization is also caused by scintillation as the refraction and dispersion cause the scattering of GNSS signals.

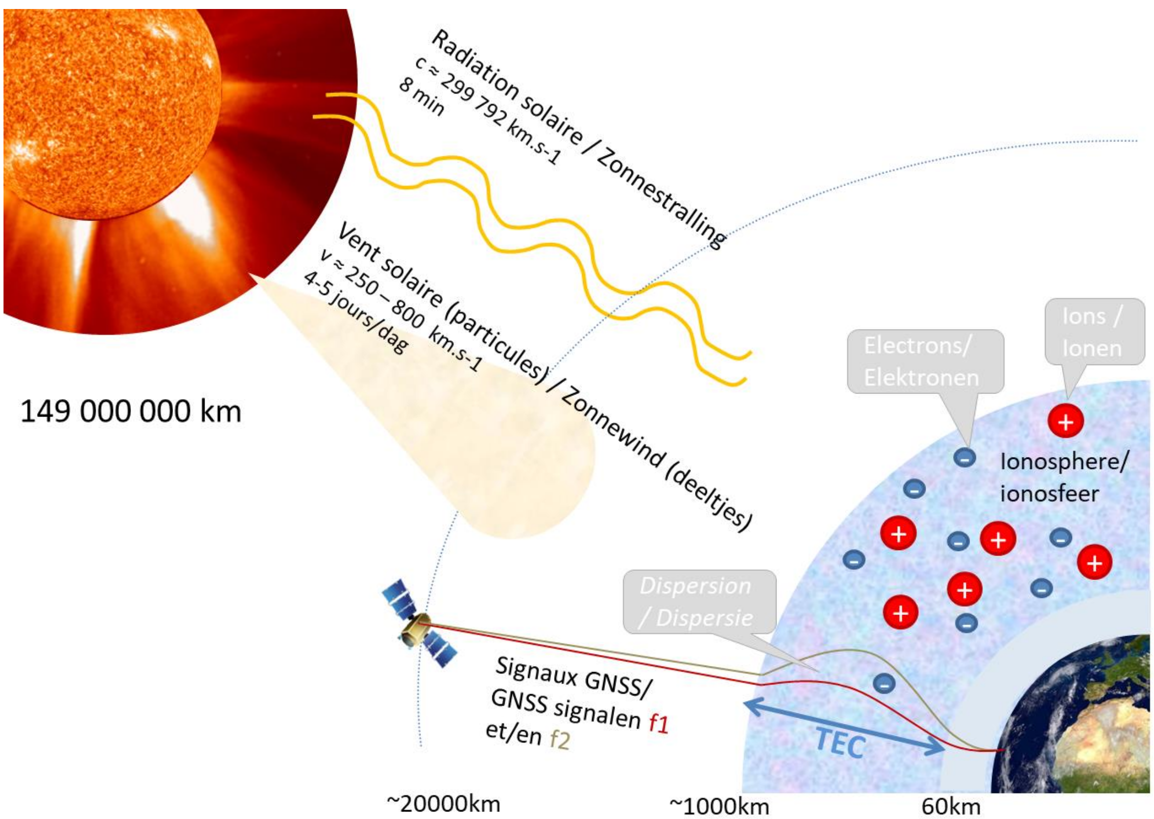

Fig. 2.2. Impact of the ionosphere on GNSS signals [Courtesy: Royal Observatory of Belgium]

The impact of the ionosphere layer on the GNSS signals has been widely investigated by various researchers [32] [33]. The Sun can provide its energy in the form of EMR to the ionosphere, which can excite free electrons and initiate the ionization process in the ionosphere, hence the ionosphere tends to disrupt GNSS signals [34]. However, several models and methods are practiced by researchers to mitigate different ionospheric scintillations and to achieve maximum accuracy from the GNSS constellations.

2.5. Seismic Anomalies Associated with Atmosphere

In this chapter so far, the impact of the ionosphere on GNSS signals, solar and geomagnetic storms along their sources have been discussed. This section discusses the behavior of the atmosphere during the pre-seismic and post-seismic events. Furthermore, analysis of the various atmospheric parameters that may impact the atmosphere during the seismic activity will be discussed in this thesis. As “Earth Observational Satellites” and “GNSS” hover over the Earth, they can help us by providing the requisite information regarding the behavior of the Earth’s atmosphere. Several studies suggest GNSS may provide information on the ionospheric disturbances before the impending EQ [35] [36] [37] and [38]. Studies also suggest that atmosphere can play a very vital role in the forecasting of an EQ [39][40]. Recent researchers try to find a mechanism to associate abnormal behavior of the atmosphere during seismic activity. Freund [41] proposed that the holes emitting from the lithosphere interact with water vapors in the lower atmosphere causing the condensation process which eventually drifts the charged particles vertically drift upwards and disturbs TEC in the ionosphere. Adil et al., [42] describe the Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) phenomena by analyzing the emanation of the radon from the Earth’s crust. Radon interacts with the water molecule in the troposphere to start the condensation process which enhances the air temperature and thermal anomalies. As a result, electron vertically drifts upwards into the ionosphere and causes disturbances in the ionospheric TEC. Fig. 2.7. illustrates a chain mechanism of LAIC phenomena. Qin et al., [43] examines aerosol behavior and variations before seismic activity and propose the enhancement of aerosol particle before the impending EQ. Several studies also suggest increase air temperature and increase in soil moisture content due to an increase in convection heat flux due to enhancement of radon in the atmosphere [44] [45].

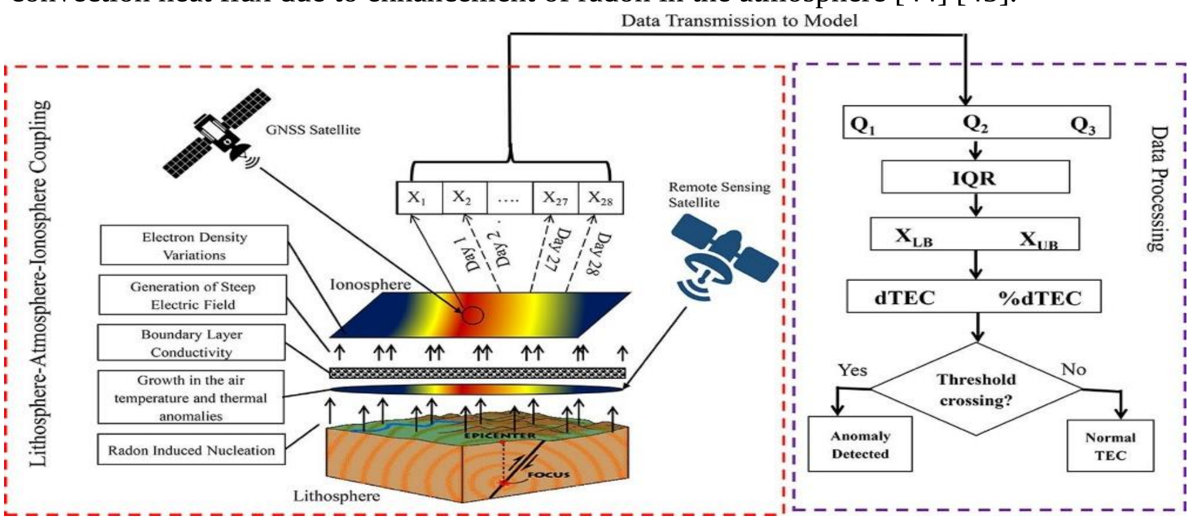

Fig. 2.3. The emanation of radon gas and LAIC phenomena [Courtesy: Adil et al., 2021]

This chapter discusses the Earth's atmosphere morphology, seismic induce anomalies, and their relationship with atmosphere behavior during seismic activity. In the next chapter, data extraction and methodology of our findings regarding Aerosol Optical Depth (AOD) and Soil Moisture ratio will be covered.

3. Data Extraction and Methodology

3.1. Case Study

In the thesis, ionospheric and atmospheric conditions before and after the two EQs from Afghanistan ($M_w > 6.0$), one EQ from the Tajikistan region ($M_w > 6.0$) and one EQ of M_w=6.1 in the El Salvador region are examined. On October 25, 2005, a magnitude of 7.0 EQ hit the Afghanistan region. Likewise, on October 05, 2008, another EQ of 6.0 M_w hit the Afghanistan region also on January 24, 2011, a magnitude of 6.0 seismic activity was observed in the Tajikistan region. Similarly, a strong EQ of M_w=6.1 observed around the El Salvador region on October 28. 2018. United States Geological Survey (USGS) provides EQ information along with its intensity. The different parameters of the EQs are listed in Table. 3.1. We studied four different hypo-central depth EQs (e.g. shallow, intermediate, and deep) on the mid-latitude part of the lithosphere in Afghanistan, Tajikistan and El Salvador (Fig. 3.1). To examine seismic behavior, it is pertinent to determine the radius or D region of the EQ. Therefore, the EQ preparation zone area is calculated by the Dobrovolsky formula [46].

$$\rho = 10^{0.43M} \qquad (3.1)$$

Where, ρ donates the EQ preparation zone in km and M represents the radius of the EQ.

3.1.1. LAIC Mechanism

This thesis strengthens the argument of LAIC phenomena to analyze and predict the seismic associated atmospheric and ionospheric anomalies. In the past, each parameter was examined separately which usually requires a lot of time and effort, this mechanism links different phenomena and formed a chain mechanism that considers all the parameters associated with the seismic activity (i.e., geochemical, atmospheric, and ionospheric parameters). Therefore, by the use of this coupling mechanism, prediction of the anomaly not only makes sense but as well as able us to understand different parameters with in-depth knowledge. Researchers use different parameters to analyze seismic precursors and their impact on the lower and upper atmosphere [36] [41] [42] [47]. Shah et al., [48] investigate seismo-ionospheric anomalies using GPS-TEC, f_oF2, Outgoing Longwave Radiation (OLR), and geo-potential height. He proposed, during the seismogenic period the radon emanating from the lithosphere causes disturbances in the ionosphere, furthermore, the air temperature and geo-potential height can also increase during the seismogenic period within the EQ breeding zone. Similarly, Shah et al., [49], proposed a LAIC model based on previous research findings fig. 3.1. According to them, the sudden release of the radon gas from the hypocenter drifts upward into the troposphere where these gas excite the ion particles causes disturbances in the upper atmosphere (i.e., ionosphere).

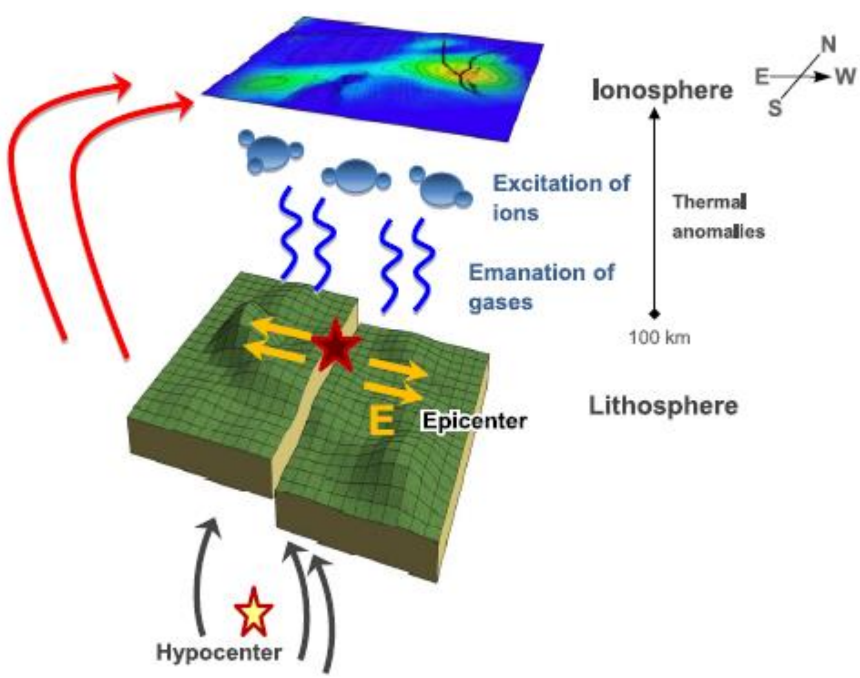

Fig. 3.1. Demonstration of LAI coupling model [Courtesy: Shah et al., [49]]

Likewise, Freund., [41] and Kuo et al., [50] proposed that the stress formed within the rocks causes the emanation of the positively charged carriers (p hole) from the stressed rocks in the lithosphere vertically drifts upwards due to current dynamos that in results may change the TEC from the ionosphere in their respective LAI coupling models (fig. 3.2).

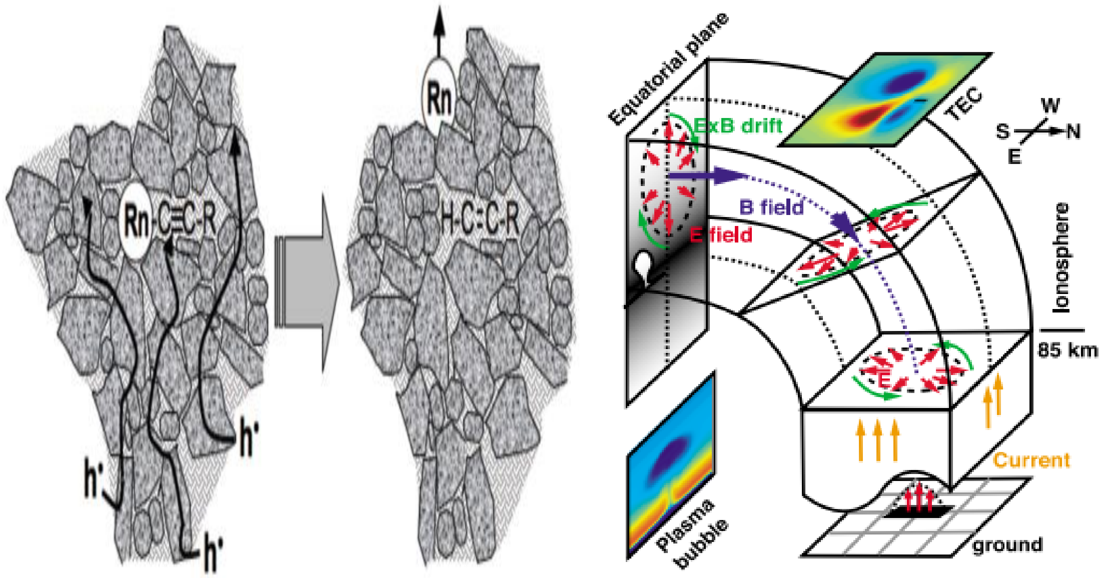

Fig. 3.2. (a) shows the emanation of p holes from the stressed rock during the seismogenic period. Whereas (b) illustrates the current dynamos due to the release of stress in the form of p holes [Courtesy: Freund., [41] and Kuo et al., [50]]

Furthermore, Sorokin and Hayakawa., [51] in their atmospheric-ionospheric coupling model explained that the emission of radon particles from the lithosphere and aerosol particles that emit from the soil gases particles reacts with the lower atmosphere and causes disturbances in the Near-ground atmosphere. They proposed that the reaction of radon and aerosol with the lower atmosphere generates an electric field that can impact the atmosphere as well as the ionosphere. Pulinets., [52] presented his LAI coupling model that is based on the emanation of radon gases from the stressed region during the seismic period released from the lithosphere into the near-ground atmosphere and reacts with water vapors in the lower atmosphere that in results causes drop in relative humidity in the atmosphere. As a result of this sudden drop in the humidity, the air temperature rises which may cause anomalies in the OLR. Pulinets., [52] further explained that the ions hydration due to decaying of the radon in the atmosphere causes changes in the air conductivity which leads to generating atmospheric current, and hence this atmospheric current drifts vertically upward into the upper atmosphere and is the main cause of the have abnormalities into the ionosphere and TEC. Fig. 3.3. explains the LAIC model proposed by Pulinets [52].

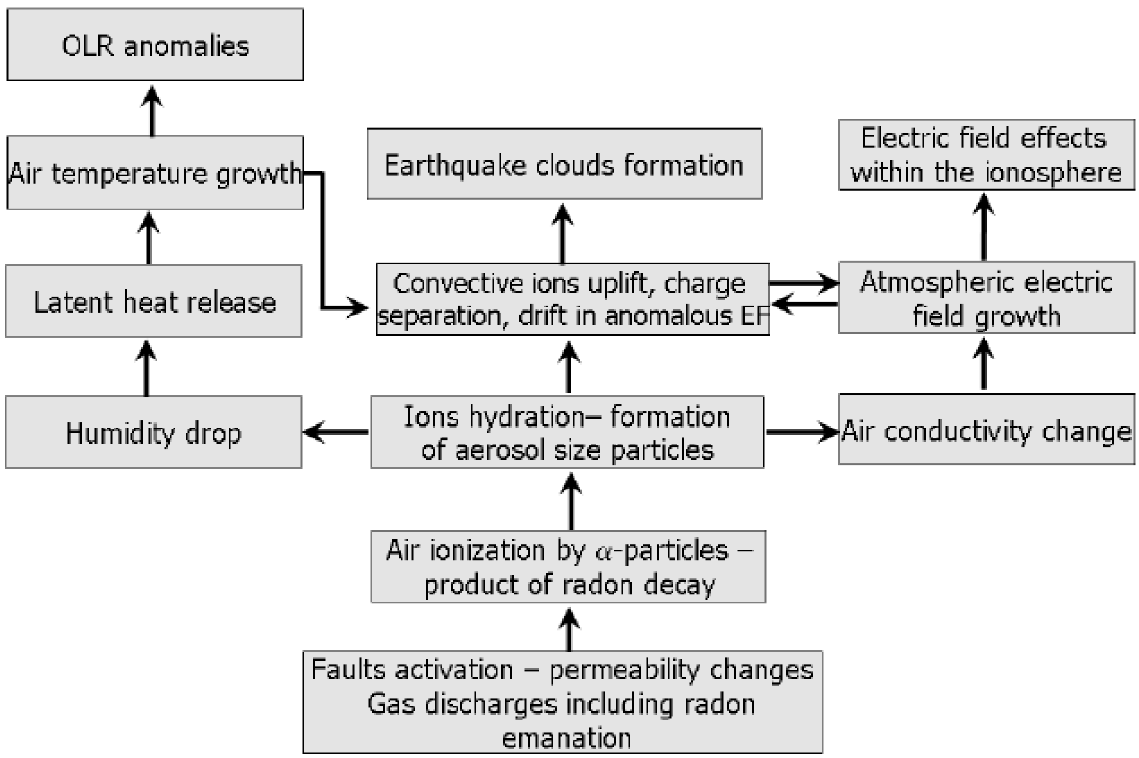

Fig. 3.3. Algorithm of the proposed LAIC model presented by Pulinets., [52]

3.1.2. Statistical Method

To analyze EQ-related disturbances statistical approach is made. Confidence Bounds are utilized to investigate the Soil Moisture, AOD, TEC, and humidity data. To identify any significant deviation triggered by the EQ, all the variables were studied for 3-4 months before and one month after the main shock. Furthermore, the anomalies were detected using μ±σ (median ± sigma), which is obtained from the data (2002-2018). To examine these parameters, long-term data (3 months before and one month after) is analyzed. Confidence intervals are used as follows:

$$UB = \tilde{X} + IQR \tag{3.2}$$

$$LB = \tilde{X} - IQR \tag{3.3}$$

Where UB, and LB denotes upper bound and lower bound respectively. $\tilde{X}$ is the Median of all the data and IQR is known as the interquartile range.

3.1.3. Digital Elevation Model

In this thesis, Digital Elevation Model (DEM) data are used to analyze the elevated terrain during the rupture caused by an EQ. The purpose of DEM is to monitor the surface deformation caused by cracks of EQ, and tilt associated the seismic activity. The temporal data used in this study, DEM data is retrieved from the National Aeronautics and Space Administration (NASA) via earth explorer (https://earthexplorer.usgs.gov/).

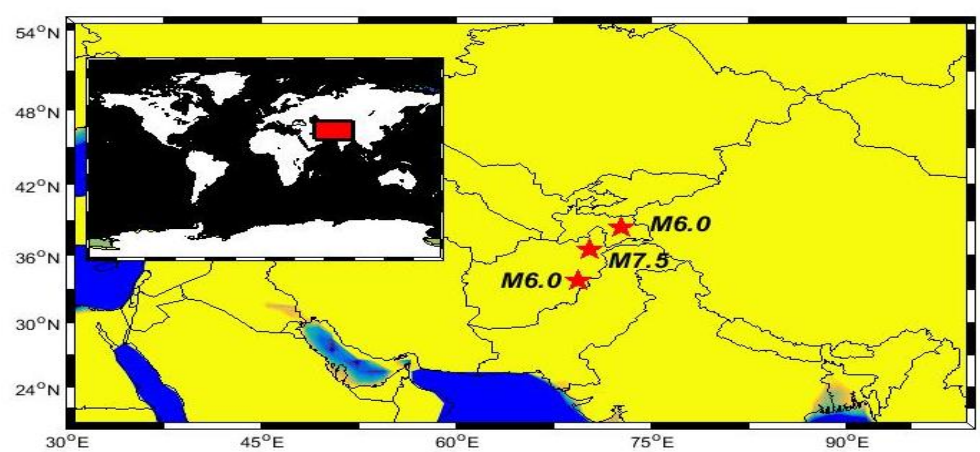

Fig. 3.4. EQs magnitudes along with their geographical location

Similarly, the Geoid height and Gravity anomaly are obtained from the ICE (https://www.aviso.altimetry.fr/es/applications/geodesy-and-geophysics/geodesy.html). In this study, we aim to show the consistency in atmospheric anomalies before the continental EQs, which occurred on high terrain in Afghanistan and Tajikistan.

3.1.4. Atmospheric Parameters

Atmospheric parameters are used to monitor atmospheric behavior during seismic activity. Soil Moisture content is analyzed to monitor the release of energy from the lithosphere into the atmosphere. This release of energy is the cause of variation in the air temperature.

Table 3.1. Anomalies Associated with Mw>6.0 EQs

Earthquake	**Occurrence Date**	**Pre-seismic anomalies days**	**Post-seismic anomalies days**
M6.0 Afghanistan	2008-10-05	-35 days before	Nil
M6.0 Tajikistan	2011-01-24	-50 days before	Nil
M7.0 Afghanistan	2005-10-25	-5 days before	5 days after
M6.1 El Salvador	2018-10-28	-2 days before	Nil

Likewise, relative humidity and AOD are also analyzed to monitor the behavior of the EQ-affected region during the impending EQ. The atmospheric data is retrieved from a NASA-based data center (http://www.giovanni.sci.gsfc.nasa.gov/giovanni). GIOVANNI is an online data source of remote sensing to assist in atmospheric and meteorology research and is maintained by NASA.

3.1.5. Total Electron Content

In this thesis, TEC is extracted to analyze the ionospheric irregularities before the impending EQ. The purpose of TEC is to monitor the ionospheric disturbances associated with seismic activity before the impending of an EQ. The GNSS-TEC data used in this study is retrieved from the IONOLAB-based software that uses RINEX files and calculates biases to extract TEC data (http://www.ionolab.org/).

Slant TEC (STEC) from the GNSS ground stations is obtained along the ray path from the satellite to the receiver [53][54]. To examine vertical TEC (VTEC), STEC has been extracted from IGS ground stations from the equation.

$$STEC_a^h = -\frac{f_1^2 f_2^2}{40.3\left(f_1^2 - f_2^2\right)}\left[P_{4,a}^h - c\ (DCB_a + DCB^h)\right] \quad (3.4)$$

From the above (2), f_1 and f_2 are the GPS dual frequencies, c is the speed of light, $P_{4,a}^h$ are the differences between smoothed code measurements while DCB_a and DCB^h are differential code biases for the satellite and receiver [55]. Furthermore, to convert STEC into VTEC following equation has been applied to STEC:

$$VTEC = STEC\ x\ cos\left[arcsin\left(\frac{Rsinz}{R+H}\right)\right] \quad (3.5)$$

In this equation, H is the height of the top of the ionospheric layer, R is the radius of Earth, and z is the zenith angle of the satellite [56]. The station-based TECs are calculated by weighting or surface interpolation of vTECs [31].

$$TEC = {\sum w(\varepsilon) VTEC} \Big/ {\sum w(\varepsilon)} \quad (3.6)$$

TEC is represented in the TEC unit (1 TECu is equal to $10^{16} \mathrm{el/m^2}$). The TEC from every station within the EPA is retrieved by the [54] method and the continuous observations are scanned for 10 days before and 5 days after the mainshock.

4. Result and Discussion

In this study, we showed the ionospheric and atmospheric anomalies before four large magnitudes EQs from GNSS-TEC and other remote sensing data in the mid-latitude of the Afghanistan, El Salvador and Tajikistan. The occurrence of anomaly over the epicenter throughout the EQ preparation time in humidity, AOD, GPS-TEC and Soil moisture before the main shock confirmed the synoptic blocking of huge energy appeared in remote sensing procedures. There exist different conclusions in past about the formation of such anomalies at the LAIC interface over the seismic breeding zone.

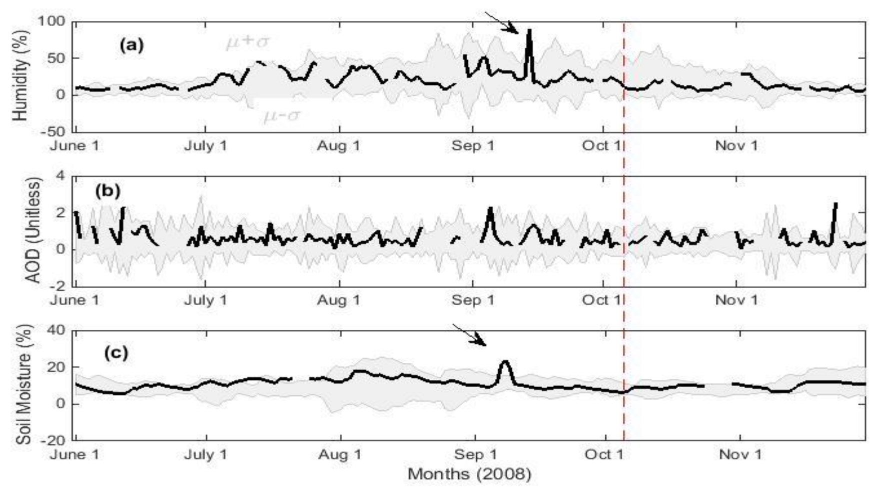

Fig. 4.1. The analysis of different atmospheric indices with their confidence bounds before and after the 5th October 2008 Afghanistan EQ.

The atmospheric anomalies are clear on the day (-35) before the 2008 Afghanistan EQ (Fig. 4.1). On the other hand, there are no post-seismic atmospheric anomalies after the main shock day within the seismogenic zone. The 2011 Tajikistan EQ shows clear anomalies in AOD on -50 days as a pre-seismic atmospheric perturbation (Fig. 4.2). This is duly verified by the analyses of humidity and soil moisture from different remote sensing satellite data (Fig. 4.2).

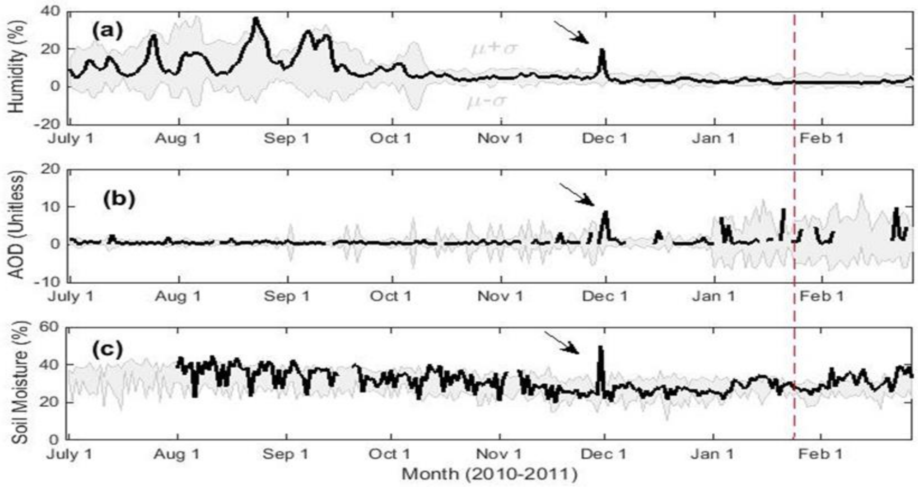

Fig. 4.2. (a) Humidity, (b) AOD, and (c) Soil Moisture analyses with confidence bounds associated with the 24th January, 2011 Tajikistan EQ

As in Fig. 4.3, the -5 day anomalies in the atmosphere are clear within 5 days before the 2005 M7.0 EQ in Afghanistan (Fig. 4.3). The AOD shows more than 10% variation as an anomaly from the normal distribution of the data (Fig. 4.3). Similar variations are recorded in humidity and soil moisture in Figures 4.3(a) and 4.3(c), respectively.

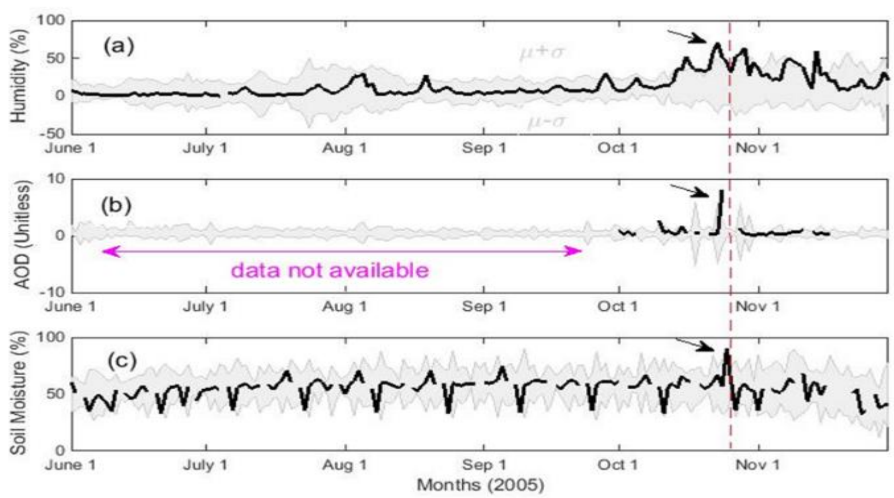

Fig. 4.3. The analysis of different atmospheric indices with their confidence bounds before and after the 25th October 2005 Afghanistan EQ

Likewise, this study also examines the morphological features associated with the El Salvador EQ that occurred on 28 October 2018 by utilizing GPS-TEC dataset. The ionospheric variations were monitored by two IGS stations falling within the EPA named GUAT and MANA. The temporal TEC time series for 10 days before and 5 days after the EQ day, along with the dTEC values, are represented in Fig. 4.4. Significant anomalies in the terms of dTEC can be observed between 22-26 October 2018 (2-6 days before the EQ), where a deviation of about 3 TECu can be observed at both stations, simultaneously. In another study, [57] inspected the morphological parameters such as synchronized TEC and OLR variances within 5 days during the analysis of M 9.0 Tohoku EQ. [58] detected approximately comparable decrements and enhancements in ionospheric TEC as possible EQ forerunners using a solid system of GPS receivers around the neighborhood of

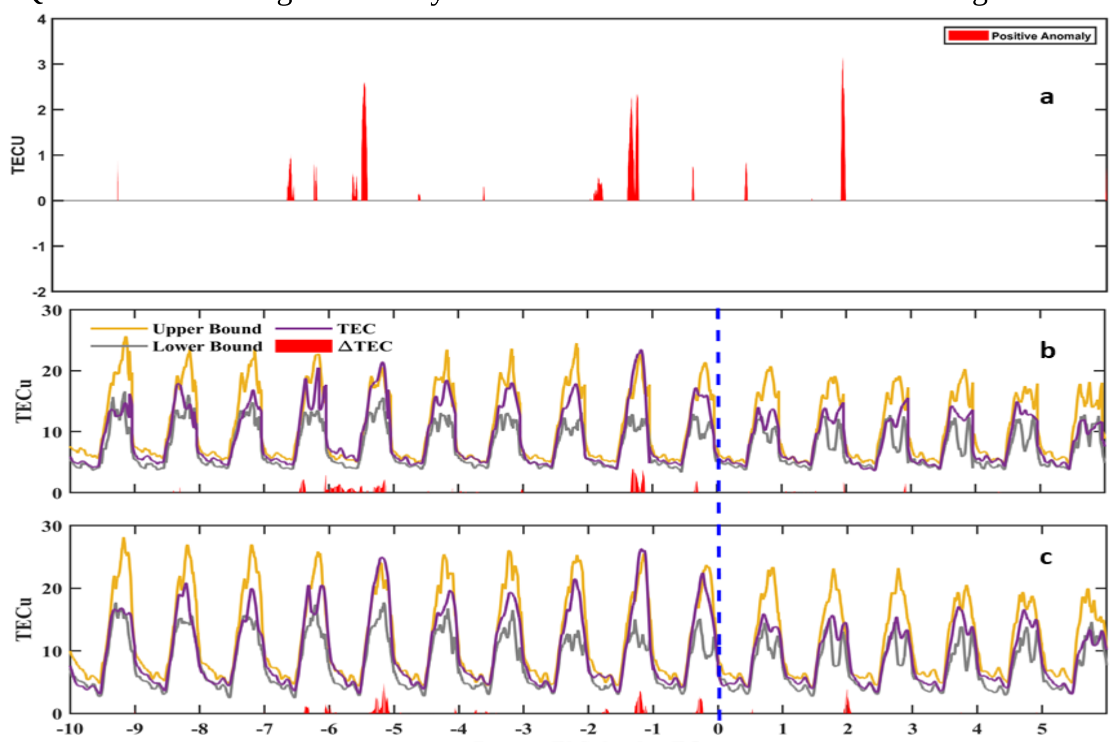

Fig. 4.4. Analysis of TEC and dTEC values recorded at (a) Quantifies the TEC using dTEC analysis (b) GUAT and (c) MANA stations, respectively, analyzed TEC and dTEC data for 10 days before and 5 days after the EQ

Wenchuan EQ. Atmospheric parameters are also spatially analyzed over the Afghanistan region for the month of September (i.e., EQ month) to visualize the abnormalities in the Humidity, AOD, and Soil moisture content. The result indicates the sufficient anomalies in the atmospheric parameters over the epicenter region of the seismogenic region as visualize in the fig. 4.5. Hence, we suggest that mutual ionospheric-atmospheric analysis may provide significant contributions to future EQ forecasting.

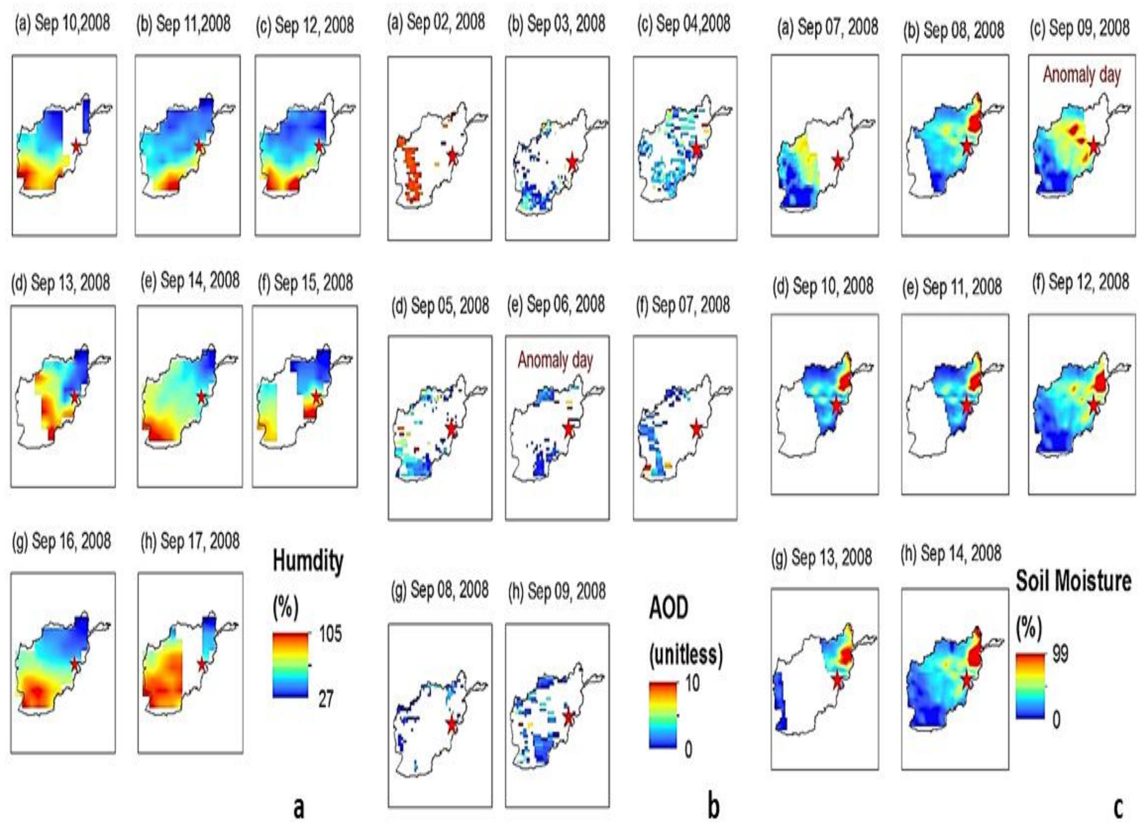

Fig. 4.5. Spatial analysis of atmospheric parameters. Panel (a) indicates the Humidity, panel (b) shows AOD, while panel (c) represents soil moisture

5. Conclusion

In this thesis, we study atmospheric anomalies associated with 4 large magnitude EQs in the mid-latitude region; i.e., Afghanistan, El Salvador and Tajikistan. The ionospheric TEC observations were treated under a statistical analysis based on confidence bounds where considerable deviations of 2 to 3 TECu were observed on October 23-26, 2018. All the variables were studied for 3-4 months before and one month after the main shock. Furthermore, the anomalies were detected using $\mu \pm \sigma$ (median $\pm$ sigma), which is obtained from the data (2002-2018). TEC anomalies, may be attributed to the increased radon emanations around the El Salvador EQ preparation zone that consequently disturbed the ionosphere. Interestingly, the anomalous response in the atmospheric-ionospheric parameters was enhancing concerning days to the EQ. We found different ionospheric and atmospheric parameters behaved abnormally over the seismic zone within several days to several months before the impending main shock. The occurrence anomaly over the epicenter during the EQ preparation period in humidity, AOD, GPS-TEC and Soil moisture content before the main shock confirmed the synoptic blocking of huge energy appeared in remote sensing procedures. We declared the anomaly as the deviation from normal distribution to a significant level beyond the confidence levels of high terrain at the lithosphere-atmosphere-ionosphere interface over the seismic breeding zone.

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